A Comprehensive Blueprint for Modeling the Japan Power Market

I. The Japanese Power Market: A Modeler's Primer

A robust fundamental model of an electricity market must be a direct reflection of its physical and regulatory architecture. The Japanese power market is characterized by unique geographical constraints and a multi-layered institutional framework that dictates dispatch and price formation. A thorough understanding of this context is a prerequisite for developing an accurate simulation model.

1.1. Deconstructing the Zonal Market: Physical Constraints and Topography

The physical structure of the Japanese grid is the primary determinant of power flow and regional price dynamics. Unlike more interconnected systems, Japan's grid is best understood as a series of semi-isolated zones linked by critical, and often constrained, transmission pathways.

Nine Interconnected Regions

The national grid, excluding the island of Okinawa, is composed of nine distinct regional grids, each historically operated by a vertically integrated utility, or Electric Power Company (EPCO). These regions are: Hokkaido, Tohoku, Tokyo, Chubu, Hokuriku, Kansai, Chugoku, Shikoku, and Kyushu.[1] Any credible market model must adopt this zonal structure as its fundamental spatial resolution. A national "copper plate" model, which assumes no internal transmission

congestion, would fail to capture the most salient features of the Japanese market.

The 50/60Hz Frequency Divide

The most significant and defining characteristic of the Japanese power system is its division into two asynchronous frequency zones.[1] Eastern Japan, encompassing the Hokkaido, Tohoku, and Tokyo regions, operates at a frequency of 50 Hz. Western Japan, which includes the Chubu, Hokuriku, Kansai, Chugoku, Shikoku, and Kyushu regions, operates at 60 Hz. This historical artifact, stemming from the procurement of German AEG generators for the East and American General Electric generators for the West in the late 19th century, creates a fundamental barrier to the free flow of alternating current (AC) power between the two halves of the country.[3]

This frequency divide is not merely an operational curiosity; it is a hard physical constraint that bifurcates the national market. The model must explicitly recognize this boundary, as it dictates which regional power exchanges are physically possible via standard AC transmission and which require specialized conversion.

Interconnections and HVDC Converters

Power transfer between the nine regions occurs via AC inter-regional lines, but more critically, power transfer between the 50 Hz East and 60 Hz West is only possible through a limited number of High-Voltage Direct Current (HVDC) frequency converter stations.[1] These stations convert AC power to DC, transmit it across the frequency boundary, and then convert it back to AC.

The total capacity of these frequency converters is currently approximately 2,100 MW (2.1 GW), with plans for significant expansion to 3 GW by 2027 to facilitate greater renewable energy integration.[1] This capacity is minuscule relative to the peak demand of either zone, which can exceed 160 GW nationally.[1] Consequently, these HVDC links represent a major bottleneck in the national grid. They are frequently congested, leading to significant and persistent price separation between Eastern and Western Japan.

The model's transmission component must therefore represent not only the AC inter-regional transfer capacities but, most importantly, the hard capacity limits of these HVDC frequency converters. These links are the primary mechanism for national price convergence, and their constraints are a key driver of regional price volatility. The frequent divergence of regional

prices, observable in historical data from the Japan Electric Power Exchange (JEPX), is a direct consequence of these physical limitations.[5] This implies that a model based on the principles of Locational Marginal Pricing (LMP), where prices are influenced by transmission congestion, is the appropriate theoretical framework, even if Japan's market design uses a zonal pricing system. The dual variables, or shadow prices, associated with these binding transmission constraints in the optimization model will be the direct mathematical explanation for the observed price differentials between regions.

1.2. Key Institutions and Market Segments

The operation of the physical grid is governed by a set of institutions and market structures that have evolved significantly since market liberalization began in 1995.[1]

Regulatory and Coordination Bodies

Two primary organizations oversee the market:

- **Ministry of Economy, Trade and Industry (METI)**: The primary government body responsible for national energy policy, long-term strategic planning, and regulation of the electricity sector.[1]
- **Organization for Cross-regional Coordination of Transmission Operators (OCCTO)**: Established in 2015, OCCTO is a pivotal institution for any market modeler. It is responsible for nationwide grid coordination, managing cross-regional power flows, ensuring supply security, and publishing a vast array of critical system data, including historical demand, generation by fuel type, and interconnector flows.[1]

The Wholesale Market (JEPX)

The Japan Electric Power Exchange (JEPX) is the country's sole wholesale electricity exchange, serving as the primary venue for generators and retailers to trade power.[2] The model developed in this plan will focus on simulating the

Day-Ahead Spot Market. In this market, participants submit bids and offers for each of the 48 half-hour trading intervals of the following day. JEPX conducts a blind, single-price auction

to determine the market clearing price and dispatch volume for each interval in each of the nine regions.[2] The historical area prices from this market are the primary target variables that the model must be calibrated to replicate.

Ancillary and Capacity Markets

While the primary focus is the Day-Ahead energy market, other market segments influence generator bidding behavior and must be considered:

- **Balancing Market**: Launched in 2021, this market procures the real-time adjustment capacity needed to balance the grid.[1]
- **Capacity Market**: The first auction was held in 2020 for delivery in 2024. This market provides a forward revenue stream to generators for being available to produce power, ensuring long-term resource adequacy.[1] Payments are made based on available capacity (JPY/kW) rather than energy produced (JPY/kWh).

The existence of the Capacity Market is particularly important. It provides a separate revenue stream that helps generators cover their fixed costs. This addresses the "missing money" problem inherent in energy-only markets. Consequently, generators may not need to recover all their fixed costs through their energy bids in the JEPX Day-Ahead market. This can lead to bidding behavior that deviates from a simple short-run marginal cost (SRMC) calculation. The model must account for this by incorporating calibration factors that represent this strategic bidding behavior. Furthermore, METI is considering a future "Simultaneous Market" design, similar to those in the US, which would co-optimize energy and balancing services through three-part bids (startup, no-load, and energy costs).[10] While the current model will focus on the existing Day-Ahead market, this future evolution should be considered in the model's architectural design.

II. Architecting the Model's Data Foundation

Building a robust power market model is fundamentally a data engineering challenge. The accuracy of the model's outputs is entirely dependent on the quality, granularity, and comprehensiveness of its inputs. Given the complexity of the Japanese market and the fragmented nature of its data sources, establishing a systematic and automated data architecture is a critical first step. This addresses the common challenge of "fighting ugly data," a sentiment expressed by analysts familiar with the market [Image 3].

2.1. Assembling the Generation Fleet Database

The static core of the model is a comprehensive database of every significant power generation unit in the nine mainland regions. This database serves as the central repository for the techno-economic parameters that define each generator's operational capabilities and costs.

- **Objective**: To create a single, authoritative table that consolidates disparate information into a structured format suitable for direct use in the optimization model.
- **Data Sources**:
  - **OCCTO**: The annual "Aggregation of Electricity Supply Plans" provides a high-level, forward-looking summary of generation capacity submitted by all utilities.[8]
  - **JEPIC**: The annual "Electric Power Industry in Japan" report offers detailed statistics on installed capacity by fuel type and technology, and often lists major power stations.[1]
  - **Regional EPCOs**: The annual reports, investor presentations, and websites of the ten regional power companies (e.g., TEPCO, KEPCO, Chubu Electric Power) are invaluable for detailed information on their specific generation assets, including plant names, capacities, and technologies.[13]
  - **Industry and Government Reports**: Publications from METI and other industry bodies often contain data on new plant commissioning and retirements.

A structured, relational database approach is essential for managing the hundreds of generating units in the model. Each parameter in the database corresponds directly to a variable or constraint in the mathematical model. For example, Capacity_MW defines the upper bound of the power output variable (pg,t), VOM_JPY/MWh is a direct component of the objective function's cost coefficient, and Min_Up_Time_hr defines an inter-temporal constraint on the commitment status variable (ug,t). This database acts as the critical bridge between raw data and the mathematical formulation.

The following table presents a recommended schema for this master database.

Table 1: Master Generation Fleet Database Schema

Field Name	Data Type	Description & Example
Plant_ID	VARCHAR	Unique identifier for each generating unit (e.g., "TEPCO_FUTTSU_CCGT_1")

Plant_Name	VARCHAR	Official name of the power station
Owner	VARCHAR	Owning entity (e.g., "TEPCO", "JERA", "J-POWER")
Region	VARCHAR	OCCTO region (e.g., "Tokyo", "Kansai")
Technology_Type	VARCHAR	Specific technology (e.g., "CCGT", "USC Coal", "ABWR Nuclear")
Fuel	VARCHAR	Primary fuel source (e.g., "LNG", "Thermal Coal", "Uranium", "Hydro")
Capacity_MW	FLOAT	Net dependable capacity in MW
Commissioning_Year	INTEGER	Year the unit began commercial operation
Efficiency_LHV	FLOAT	Thermal efficiency based on Lower Heating Value (%)
Heat_Rate_GJ/MWh	FLOAT	Energy input per unit of electricity output; inverse of efficiency
VOM_JPY/MWh	FLOAT	Variable Operations & Maintenance cost
FOM_JPY/kW/yr	FLOAT	Fixed Operations & Maintenance cost
Startup_Cost_JPY	FLOAT	Cost to bring the unit online from a cold state

Ramp_Up_MW/min	FLOAT	Maximum rate of increase in power output
Ramp_Down_MW/min	FLOAT	Maximum rate of decrease in power output
Min_Up_Time_hr	INTEGER	Minimum hours a unit must run once started
Min_Down_Time_hr	INTEGER	Minimum hours a unit must remain offline once stopped
Min_Load_MW	FLOAT	Minimum stable generation level (P_min)

2.2. Sourcing and Processing Input Time-Series Data

The model requires a suite of time-dependent inputs, ideally at a 30-minute resolution to align with the JEPX Day-Ahead market structure.

- **Regional Demand (Load)**: This is the primary exogenous driver of the model. Historical 30-minute load data for each of the nine regions is published by the respective regional Transmission System Operators (TSOs). These are typically available as downloadable CSV files from their public websites.[20]
- **Renewable Generation Profiles**: Historical 30-minute generation data for solar and wind power for each region is also published by the TSOs and aggregated by OCCTO.[20] In a dispatch model, this intermittent generation is typically treated as a "negative load," as it is must-take and reduces the amount of demand that must be met by dispatchable generators.
- **Interconnector Flows and Availability**: OCCTO publishes historical data on actual power flows on the inter-regional transmission lines.[28] This data is invaluable for validating the model's dispatch and power flow logic. Information on planned and unplanned outages of these lines is also critical for accurately modeling transmission constraints.
- **Historical Prices**: JEPX is the source for historical Day-Ahead spot prices, providing data for the national system price as well as area prices for each of the nine regions.[5] This dataset is the primary benchmark against which the model's price outputs will be

calibrated and validated.

2.3. Fuel and Carbon Price Intelligence

The marginal cost of thermal generation is a direct function of fuel and carbon prices. As Japan is heavily reliant on imported fossil fuels, tracking these commodity prices is essential.

- **Liquefied Natural Gas (LNG)**: Japan's power sector is a major consumer of LNG. The pricing is complex, as a significant portion is procured via long-term contracts linked to oil prices, while a growing share is purchased on the spot market. A blended price approach is recommended. The Japan Crude Cocktail (JCC), a weighted average of crude oil import prices, serves as a strong proxy for the oil-linked contracts.[2] This should be combined with the Japan Korea Marker (JKM), the benchmark for spot LNG in Asia. The Ministry of Finance (MOF) trade statistics provide official data on average LNG import prices, which can be used for validation.[32]
- **Coal**: The primary benchmark for thermal coal imported into Japan is the Newcastle (Australia) price, which must be adjusted for freight costs. The MOF trade statistics also provide detailed data on the volume and average price of imported thermal coal, which can be used to construct a reliable historical price series.[35]
- **Oil**: Older, less efficient oil-fired power plants are often the marginal units during peak demand periods. Prices for crude oil (such as JCC) or specific fuel oils (like high-sulfur fuel oil) are needed to model their costs accurately.[32]
- **Carbon Pricing**: Japan is implementing its "Green Transformation" (GX) policy, which includes a carbon pricing mechanism. This consists of an emissions trading scheme (GX-ETS) starting in FY2026 and a fossil fuel surcharge from FY2028.[1] The model must include a carbon cost component in the SRMC calculation. Initially, this can be modeled as a fixed adder based on government policy documents. As the market develops, this should be updated to reflect the actual traded price of carbon allowances.

To provide a clear roadmap for data acquisition, the following table maps each required data component to its primary source and key characteristics.

Table 2: Key Data Sources and Granularity

Data Component	Primary Source	URL/API Endpoint	Format	Temporal Resolution	Update Frequency
Regional Demand	Regional TSOs (e.g.,	e.g., tepco.co.jp/	CSV	30-min	Daily

(Load)	TEPCO PG)	forecast/			
Renewable Generation	Regional TSOs / OCCTO	e.g., setsuden.nw.tohoku-epco.co.jp	CSV	30-min	Daily
Interconnector Flows	OCCTO	occto.or.jp	CSV	Hourly / 5-min	Monthly
Historical Area Prices	JEPX	jepx.jp/en/market-data/spot/	CSV	30-min	Daily
LNG Prices (JCC, JKM)	Commodity Data Vendors / MOF	fred.stlouisfed.org, customs.go.jp	API / CSV	Daily / Monthly	Daily / Monthly
Coal Prices (Newcastle)	Commodity Data Vendors / MOF	tradingeconomics.com, customs.go.jp	API / CSV	Daily / Monthly	Daily / Monthly
Plant Outage Schedules	JEPX / OCCTO	jepx.jp / occto.or.jp	Web/PDF	Daily	Daily

III. Constructing the Regional Bid Stacks: Two Methodological Pillars

The supply curve, or "bid stack," is the central element of a fundamental market model. It represents the aggregated supply of electricity from all available generators at various price levels. The intersection of this supply curve with the demand for electricity determines the market clearing price. There are two primary, complementary methodologies for constructing

these regional bid stacks: a "bottom-up" approach grounded in economic first principles, and a "top-down" approach derived from observed market data.

3.1. The Economic Principles Approach: Reverse-Engineering the Merit Order

This approach, often called the "merit order" model, constructs the supply curve by calculating the Short-Run Marginal Cost (SRMC) for every available generating unit and stacking them in ascending order of cost. This represents a theoretically "rational" dispatch order, where the cheapest generators are used first.

SRMC Calculation for Thermal Generators

The SRMC is the cost to produce one additional megawatt-hour (MWh) of electricity. For a thermal power plant, it is dominated by fuel, variable O&M, and carbon costs. The core formula is:

$$SRMC_{g,t} = HR_g \times P_{fuel,t} + VOM_g + HR_g \times EF_{fuel} \times P_{carbon,t}$$

Where:

- $SRMC_{g,t}$ is the Short-Run Marginal Cost of generator g at time t (in JPY/MWh).
- HR_g is the heat rate of generator g (in GJ/MWh). This is the inverse of its thermal efficiency and is a critical parameter. For state-of-the-art plants like the Joetsu Thermal Power Station, efficiencies can reach 63.62% (LHV).[46] For older plants, standard engineering estimates must be used, such as 45-48% for Ultra-Supercritical (USC) coal plants and 36-40% for subcritical coal plants.[47]
- $P_{fuel,t}$ is the price of fuel at time t (in JPY/GJ), using the time-series data compiled in Section 2.3.
- VOM_g is the Variable Operations & Maintenance cost of generator g (in JPY/MWh). This includes costs for consumables, water, and wear-and-tear maintenance. Sourcing this data for Japan is challenging; therefore, using international benchmarks from sources like the IEA or Lazard, adjusted for local conditions, is a common starting point.[49] VOM will be a key parameter for model calibration.
- EF_{fuel} is the emission factor of the fuel (in tCO_2/GJ).
- $P_{carbon,t}$ is the price of carbon at time t (in JPY/tCO_2).

Modeling Non-Thermal Generation

Non-thermal generators have different cost structures and operational constraints:

- **Nuclear Power**: Has a very low marginal cost (primarily fuel and minor VOM). It is typically bid at or near zero JPY/MWh and treated as a baseload, must-run resource, dispatched whenever it is available. Its output is constrained by planned maintenance and any unplanned outages.
- **Renewables (Solar and Wind)**: These resources have zero marginal cost. Their generation output is determined by weather conditions, not price. In the model, their generation is treated as a price-taker, effectively reducing the "net load" that must be served by dispatchable generators.
- **Hydro (Conventional)**: While having a near-zero marginal cost, conventional hydro plants are energy-limited by their reservoir levels. Their operators aim to maximize revenue over time by dispatching water when prices are highest. A simplified but effective modeling approach is to bid them at a very low price but apply constraints on their total daily or weekly energy output based on historical generation patterns and water availability.
- **Pumped Storage Hydro (PSH)**: These are unique assets that act as both a load (during pumping) and a generator. Their operation is driven by price arbitrage. They aim to buy power (pump water uphill) during low-price periods (typically overnight) and sell power (release water to generate) during high-price periods (typically daytime peaks). Their bidding strategy is an optimization problem in itself, and they are typically dispatched by the model based on this arbitrage logic.

Synthesizing the Curve

For each 30-minute interval, the model identifies all available generators (accounting for outages), calculates their SRMC, and sorts them in ascending order. The cumulative capacity at each price point forms the regional supply curve, or merit order stack.

3.2. The Market Data Approach: Deriving Stacks from JEPX Bidding Curves

This "top-down" approach leverages the actual aggregated bidding data published by JEPX to construct the supply curve. This method captures not only the economic merit order but also the strategic bidding behaviors, risk premiums, and market power exercised by participants.

Accessing and Interpreting JEPX Data

JEPX publishes aggregated sell (supply) and buy (demand) curves for each region and trading interval.[30] This data, typically provided as a series of price-quantity pairs, forms a step-wise function representing the total volume offered at or below each price level. The shape of this curve is highly informative:

- **Flat, low-price segments**: Typically represent baseload generation with low marginal costs, such as nuclear, run-of-river hydro, and must-run coal plants.
- **Steeply sloped segments**: Represent mid-merit and peaking plants, such as CCGTs and oil-fired units, where capacity is more scarce and bid prices rise rapidly.
- **Inflection points**: Often indicate the price level at which a new class of technology (e.g., switching from coal to gas) enters the dispatch stack.[52]

Parametric Curve Fitting

Using the raw, high-resolution step-function data for every one of the 17,520 half-hour intervals in a year can be computationally burdensome. A more efficient method is to fit a continuous, parametric function to the historical bid curves. This creates a smoothed, computationally tractable representation of the market's supply behavior. Common functional forms include piecewise linear, quadratic, or hyperbolic functions. For instance, a hyperbolic function of the form $Price = a + b \times (1/(C_{total} - Q_{dispatch}))$ can effectively capture the increasing scarcity premium as dispatch approaches total available capacity. The parameters of this function can then be estimated econometrically, potentially as a function of external drivers like fuel prices or demand levels.

The economic principles and market data approaches should not be viewed as mutually exclusive. The most sophisticated models often use a hybrid methodology. The SRMC model provides a fundamentally sound basis for the *relative ranking* of generators (the merit order). However, it will almost certainly produce prices that are systematically lower and less volatile than observed JEPX prices. This is because, in reality, generators do not bid their true marginal cost; they bid strategically to maximize profit, often incorporating a markup to cover

fixed costs. The difference between the SRMC-based stack and the observed JEPX stack for any given hour represents the aggregate market markup or "alpha." A powerful calibration technique is to use the SRMC to establish the merit order and then apply a calibrated bidding multiplier to the SRMC of different generator classes (e.g., a 1.1x multiplier for CCGTs, 1.3x for oil peakers) to lift and reshape the theoretical curve to match the empirically observed one. This multiplier becomes a key calibration parameter representing the complex strategic behaviors in the market.

To illustrate the foundational calculation, the following table provides a worked example of the SRMC calculation for representative thermal plant types under a hypothetical set of market conditions.

Table 3: SRMC Calculation for Representative Plant Types

Parameter	Modern CCGT	USC Coal	Old Oil Peaker
Assumed Efficiency (LHV)	60.0%	45.0%	35.0%
Heat Rate (GJ/MWh)	**6.00**	**8.00**	**10.29**
Fuel Price (JPY/GJ)	2,000	1,200	2,500
VOM (JPY/MWh)	200	300	500
Carbon Factor (tCO2/GJ)	0.056	0.094	0.074
Carbon Price (JPY/tCO2)	3,000	3,000	3,000
Fuel Cost (JPY/MWh)	12,000	9,600	25,714
Carbon Cost (JPY/MWh)	1,008	2,256	2,284
Calculated SRMC (JPY/MWh)	**13,208**	**12,156**	**28,498**

IV. The Optimization Engine: Simulating Market Dispatch

The core of the market model is an optimization engine that simulates the JEPX day-ahead auction. This engine solves a complex mathematical problem known as the Security-Constrained Unit Commitment and Economic Dispatch (SCUC-ED) problem. The goal is to determine the least-cost schedule of generator commitment and dispatch levels that satisfies demand and respects all physical and operational constraints of the power system.

4.1. Formulating the Unit Commitment & Economic Dispatch (UC-ED) Problem

The UC-ED problem is formulated as a Mixed-Integer Linear Program (MILP), a class of optimization problems that involves both continuous and binary (0/1) decision variables. The complexity of this formulation is necessary to accurately capture the inter-temporal decisions (e.g., when to start up or shut down a plant) that are central to power system operations.[54]

Objective Function

The objective is to minimize the total system production cost over a defined time horizon (typically 24 to 48 hours). The cost is the sum of production costs, startup costs, and shutdown costs for all generators ($g \in G$) over all time periods ($t \in T$):

$$\min \sum_{t \in T} \sum_{g \in G} [C_{prod}(p_{g,t}) + C_{start}(v_{g,t}) + C_{shut}(w_{g,t})]$$

Where:

- $C_{prod}(p_{g,t})$ is the cost of producing power $p_{g,t}$, typically modeled as $p_{g,t} \times SRMC_{g,t}$.
- $C_{start}(v_{g,t})$ is the cost incurred when a generator starts up ($v_{g,t}=1$).
- $C_{shut}(w_{g,t})$ is the cost incurred when a generator shuts down ($w_{g,t}=1$).

Decision Variables

- $p_{g,t}$: Power output of generator g at time t (continuous variable, in MW).
- $u_{g,t}$: Commitment status of generator g at time t (binary variable: 1 if online, 0 if offline).
- $v_{g,t}$: Startup status of generator g at time t (binary variable: 1 if the unit starts up at the beginning of period t).
- $w_{g,t}$: Shutdown status of generator g at time t (binary variable: 1 if the unit shuts down at the beginning of period t).

Key Constraints

The optimization is subject to a set of constraints that represent the physical laws and operational limits of the power system. The parameters for these constraints are drawn directly from the Master Generation Fleet Database (Table 1).

1. Power Balance Constraint: For each region r and each time period t, the total generation within the region plus net power imports must equal the region's net load. This is the fundamental supply-meets-demand constraint.

   $$\sum_{g \in G_r} p_{g,t} + NetImports_{r,t} = NetLoad_{r,t} \quad \forall r,t$$
2. Generation Limits: A generator can only produce power if it is committed to be online. Its output must be between its minimum stable generation level (Pmin) and its maximum capacity (Pmax).

   $$u_{g,t} \cdot P_{min,g} \le p_{g,t} \le u_{g,t} \cdot P_{max,g} \quad \forall g,t$$
3. Ramp Rate Limits: The change in a generator's output from one period to the next cannot exceed its ramp-up (RUg) or ramp-down (RDg) limits. This models the physical inertia of large turbines.

   $$p_{g,t} - p_{g,t-1} \le RU_g \quad \forall g,t$$
   $$p_{g,t-1} - p_{g,t} \le RD_g \quad \forall g,t$$
4. Minimum Up and Down Time Constraints: These are critical inter-temporal constraints. Once a thermal unit is started, it must remain online for a minimum number of hours ($T_{up,g}$). Similarly, once shut down, it must remain offline for a minimum duration ($T_{down,g}$) to allow for cooling and safe restart. These constraints link the binary commitment variables ($u_{g,t}$) across multiple time periods.

   $$\sum_{i=t}^{t+T_{up,g}-1} u_{g,i} \ge T_{up,g} \cdot v_{g,t} \quad \forall g,t$$
   $$\sum_{i=t}^{t+T_{down,g}-1} (1-u_{g,i}) \ge T_{down,g} \cdot w_{g,t} \quad \forall g,t$$
5. Commitment Status Logic: These constraints link the commitment, startup, and shutdown

variables together.

$$v_{g,t} - w_{g,t} = u_{g,t} - u_{g,t-1} \quad \forall g,t$$

The inclusion of binary variables and inter-temporal constraints makes the UC-ED problem computationally intensive. A simple economic dispatch (ED) model, which lacks these features, would merely dispatch generators based on their SRMC in each hour independently. Such a model would fail to capture crucial real-world dynamics. For instance, a UC model would correctly recognize that it is more cost-effective to keep a large, inflexible baseload coal plant running at minimum load overnight, even if its SRMC is higher than the market price, to avoid incurring a very high startup cost to meet the next day's morning peak. An ED model would incorrectly shut the plant down. Therefore, the full UC-ED formulation is essential for accurately simulating a thermal-dominant system like Japan's.

4.2. Modeling Inter-Regional Power Flows

To model the interconnected nine-region system, the formulation is extended to include power flow variables and transmission constraints.

The regional power balance constraint is modified to explicitly account for flows on transmission lines:
$$ \sum_{g \in G_r} p_{g,t} + \sum_{l \in L_{in,r}} f_{l,t} - \sum_{l \in L_{out,r}} f_{l,t} = \text{NetLoad}_{r,t} \quad \forall r, t $$
Where:

- $f_{l,t}$ is the power flow on interconnector l at time t (in MW).
- $L_{in,r}$ and $L_{out,r}$ are the sets of lines flowing into and out of region r, respectively.

A new set of constraints is added to limit these flows:

- Transmission Capacity Constraints: The flow on each interconnector (both AC and HVDC) cannot exceed its maximum thermal or operational capacity ($F_{max,l}$).

  $$-F_{max,l} \le f_{l,t} \le F_{max,l} \quad \forall l,t$$

4.3. Solving the Model and Extracting Prices

The complete MILP formulation is passed to a specialized solver. Solvers like Gurobi, CPLEX,

or open-source alternatives such as HiGHS and CBC use sophisticated algorithms (e.g., branch-and-cut) to find the optimal solution.

The most important output, aside from the dispatch schedule itself, is the system of electricity prices. In linear programming, every constraint has an associated dual variable, or "shadow price." The shadow price represents the change in the objective function (total system cost) for a one-unit relaxation of the constraint. The dual variable on a region's power balance constraint therefore represents the cost of supplying one additional MWh of electricity to that region. This is the economic definition of the Locational Marginal Price (LMP). The model's simulated area price for each region r at time t is the dual variable ($\lambda_{r,t}$) of the power balance constraint for that region and time.

V. Calibration, Validation, and Back-testing

A theoretical model, no matter how sophisticated, is of little practical use until it has been rigorously calibrated and validated against historical reality. The calibration process tunes the model's uncertain parameters to ensure its outputs—dispatch and prices—accurately replicate observed market behavior. This transforms the model from an academic exercise into a reliable forecasting and analysis tool.

5.1. The Calibration Workflow

Calibration is an iterative process of running the model, comparing its output to historical data, and adjusting its input parameters to minimize the discrepancy.

1. **Define a Calibration Period**: Select a historical period for which a complete and clean dataset is available. A full calendar year is ideal as it captures all seasonal variations in demand, hydro conditions, and renewable output. This is the "in-sample" period.
2. **Establish Initial Parameters**: Populate the model with initial, literature-based or engineering-based estimates for all parameters. This includes thermal efficiencies, VOM costs, startup costs, and ramp rates from the Generation Fleet Database. Initially, assume rational bidding (i.e., bidding at SRMC, or an "alpha" multiplier of 1.0).
3. **Execute the Initial Model Run**: Solve the UC-ED model for the entire calibration period.
4. **Calculate Error Metrics**: Quantify the deviation between the model's simulated results and the historical actuals using a set of Key Performance Indicators (KPIs), as detailed in Section 5.2.
5. **Systematic Parameter Adjustment**: Identify the most uncertain and impactful

parameters and adjust them to reduce the error. This is a multi-dimensional optimization problem in itself. Key parameters to adjust include:
    - **Availability/Outage Assumptions**: The model needs to account for unplanned (forced) outages. A common approach is to apply a de-rating factor to the capacity of thermal fleets based on historical availability rates.
    - **Bidding Multipliers ("Alphas")**: As discussed in Section III, generators rarely bid their true SRMC. Introduce bidding multipliers for different classes of technology (e.g., αCCGT, αCoal, αOil) that scale their SRMC-based costs. These alphas capture strategic behavior and are among the most powerful calibration levers.
    - **Cost Components**: Fine-tune the initial estimates for Variable O&M and startup costs, which are often poorly documented publicly.
    - **Hydro Dispatch Constraints**: Adjust the daily or weekly energy limits for conventional hydroelectric plants to better match their historical dispatch patterns.
6. **Iterate and Converge**: Re-run the model with the adjusted parameters and repeat the process. This can be done through manual, expert-driven adjustments or by employing automated optimization algorithms (e.g., a genetic algorithm or grid search) that seek to find the set of parameters that minimizes a composite error score. The process continues until the model's performance on the KPIs is deemed satisfactory.

5.2. Key Performance Metrics (KPIs)

The quality of the calibration is measured against a set of objective metrics that compare simulated and historical outcomes.

- **Price Accuracy**: This is often the primary goal.
    - **Root Mean Square Error (RMSE)**: Measures the magnitude of the price errors. $RMSE = \frac{1}{N}\Sigma_{t=1}^{N}(P_{sim,t} - P_{hist,t})^2$.
    - **Mean Absolute Percentage Error (MAPE)**: Measures the percentage deviation, which is useful for understanding relative error. $MAPE = \frac{1}{N}\Sigma_{t=1}^{N}\left|\frac{P_{hist,t} - P_{sim,t}}{P_{hist,t}}\right|$.
    - **Price Correlation**: The Pearson correlation coefficient between the simulated and historical price series.
- **Dispatch Accuracy**: Ensure the model is getting the right price for the right reasons.
    - **Generation by Fuel Type**: Calculate the RMSE or total absolute error for the dispatched volume (in MWh) of each major fuel type (LNG, coal, nuclear, hydro) on a daily or weekly basis for each region.
- **Interconnector Flow Accuracy**: Validate the model's handling of transmission constraints.
    - **Flow RMSE**: Calculate the RMSE between simulated and historical flows on the key inter-regional tie-lines, especially the critical HVDC converters.
- **Volatility and Distributional Metrics**:

    - **Standard Deviation**: Compare the standard deviation of simulated prices to historical prices to ensure the model captures the correct level of market volatility.
    - **Price Duration Curve**: Compare the shape of the simulated and historical price duration curves to check for systematic biases (e.g., consistently under-predicting peak prices).

5.3. Out-of-Sample Validation

The most critical test of a model's predictive power is its performance on data it has not seen before. Overfitting—tuning the model so perfectly to the calibration period that it loses its ability to generalize—is a significant risk.

- **Process**: After the calibration is finalized using the "in-sample" data (e.g., calendar year 2022), the model is run, without any further parameter changes, on a separate, "out-of-sample" period (e.g., calendar year 2023).
- **Evaluation**: The same set of KPIs is calculated for this out-of-sample period. A well-calibrated, robust model should exhibit performance on the out-of-sample data that is comparable to, though typically slightly worse than, its performance on the in-sample data. A significant degradation in performance indicates that the model has been overfitted and may not be reliable for forecasting.

VI. A Blueprint for Implementation: Code Structure and Best Practices

Translating the complex methodologies described above into functional software requires a disciplined approach to code architecture and technology selection. A modular, well-documented codebase is essential for managing the project's complexity, facilitating collaboration, and enabling future extensions.

6.1. Suggested Code Architecture

A modular design isolates different functionalities of the project, making the system easier to develop, test, and maintain. The project should be organized into a series of distinct Python

modules or packages:

1. **data_ingestion**: This module contains all scripts responsible for interacting with external data sources. It should include functions to download, parse, and clean raw data from JEPX, OCCTO, regional TSOs, and commodity price providers. The output of this module should be standardized data structures (e.g., Pandas DataFrames) stored in an intermediate data lake (e.g., a directory of Parquet or CSV files).
2. **database_builder**: This module is responsible for constructing the static datasets, primarily the Master Generation Fleet Database (Table 1). It will read from various static sources (e.g., manually curated spreadsheets, PDF reports) and produce a clean, analysis-ready database file.
3. **model_formulation**: This is the scientific core of the project. It contains the functions that build the mathematical optimization model using a high-level modeling library like Pyomo or PyPSA. This module should define the decision variables, objective function, and all constraints in a solver-agnostic manner.
4. **scenario_manager**: This module prepares the specific inputs for a given model run. It takes high-level instructions (e.g., a date range, a demand forecast scenario, a fuel price scenario) and assembles the necessary time-series data and generator parameters into the format required by the model_formulation module.
5. **solver_engine**: This is a relatively simple but crucial module. It takes the formulated model object and the scenario data, passes them to the selected MILP solver (e.g., Gurobi), executes the optimization, and retrieves the results (variable values and duals).
6. **results_analysis**: This module processes the raw output from the solver. It calculates the KPIs defined in Section 5.2, generates plots and summary tables (such as those seen in the user's initial analysis [Image 3]), and saves the results in a structured format for further analysis.
7. **calibration_tools**: This is an advanced module that automates the calibration workflow. It can implement algorithms (e.g., grid search, Bayesian optimization) to systematically vary the calibration parameters (like bidding alphas) and re-run the model to find the set of parameters that minimizes the error metrics calculated by the results_analysis module.

6.2. Recommended Technology Stack

The choice of technology should prioritize productivity, performance, and access to a robust ecosystem of libraries for data science and optimization.

- **Programming Language**: **Python** is the unequivocal choice for this type of project. Its dominance in the data science community ensures access to a vast array of high-quality, well-documented libraries.
- **Core Libraries**:
    - **Data Handling**: **Pandas** and **NumPy** are essential for manipulating the large

time-series datasets and numerical parameters involved.
    - **Optimization Modeling**: **PyPSA (Python for Power System Analysis)** or **Pyomo** are excellent high-level packages for formulating the UC-ED model in Python. PyPSA is particularly recommended as it is specifically designed for power systems and has built-in data structures and functions for handling generators, lines, and loads, which can significantly accelerate development.[54]
    - **MILP Solver**: A high-performance MILP solver is critical. While open-source solvers like **HiGHS** or **CBC** are good for initial development, solving a full-scale, multi-region UC problem for an entire year will likely require the performance of a commercial solver like **Gurobi** or **CPLEX**. Academic licenses for these are often available.
    - **Visualization**: **Matplotlib**, **Seaborn**, and **Plotly** provide comprehensive capabilities for generating static and interactive plots to analyze model results.
    - **Statistical Analysis**: **Scikit-learn** and **Statsmodels** are useful for any statistical tasks, such as fitting parametric functions to the JEPX bid curves.

6.3. Pathways to Model Extension

Once the core Day-Ahead market model is built and calibrated, its modular architecture allows for straightforward extension to investigate more complex and forward-looking questions.

- **Modeling Battery Storage**: Introduce battery energy storage systems (BESS) as a new asset class. This involves adding new variables for state-of-charge, charging, and discharging, along with new constraints for energy balance, charge/discharge efficiency, and power/energy capacity limits.
- **Incorporating Demand Response**: Model a portion of the load as price-elastic or dispatchable demand response. This can be represented as a generator with a negative cost (i.e., a willingness to pay to avoid consumption) that can be dispatched up to a certain potential.
- **Long-Term Scenario Analysis**: The calibrated model becomes a powerful tool for forecasting. By replacing historical inputs with future projections for fuel prices, carbon prices, demand growth (e.g., from data centers [4]), and changes in the generation mix (e.g., renewable expansion and nuclear restarts as outlined in Japan's 7th Strategic Energy Plan [1]), the model can simulate future electricity prices and dispatch patterns, providing valuable insights for investment and policy analysis.

VII. Conclusion and Strategic Recommendations

Modeling the Japanese power market is a formidable but achievable undertaking. The market's unique physical characteristics—notably its nine-zone structure and the 50/60 Hz frequency divide—coupled with its evolving regulatory landscape, demand a sophisticated and data-intensive approach. A simplified, national-level model would fail to capture the essential dynamics of regional price separation and transmission congestion that are fundamental to this market.

This report has laid out a comprehensive, end-to-end blueprint for constructing a robust fundamental model. The recommended approach is centered on a multi-regional Unit Commitment and Economic Dispatch (UC-ED) optimization, formulated as a Mixed-Integer Linear Program. This methodology is the industry standard for accurately simulating thermal-heavy power systems, as it correctly captures the complex inter-temporal constraints and economic trade-offs inherent in generator startup and shutdown decisions.

The critical path to success involves two parallel efforts:

1. **Systematic Data Architecture**: The challenge of acquiring and structuring the vast and often fragmented data from sources like OCCTO, JEPX, and the regional TSOs cannot be underestimated. A disciplined approach, centered on building a master generation fleet database and automated pipelines for time-series data, is paramount.
2. **Hybrid Bid Stack Construction**: The most accurate representation of the supply curve will be achieved through a hybrid methodology. A "bottom-up" merit order stack, based on the Short-Run Marginal Cost of each generator, should form the foundation. This must then be augmented and calibrated against the "top-down" reality of observed JEPX bid curves, using bidding multipliers to account for the strategic behaviors and risk premiums of market participants.

Ultimately, the model's value is determined by its ability to replicate history. A rigorous calibration and validation process, using clear Key Performance Indicators for price, dispatch, and interconnector flows, is non-negotiable. By systematically tuning the model's uncertain parameters against a historical "in-sample" period and validating its performance on an "out-of-sample" period, the analyst can build confidence in the model's predictive power.

By following the structured plan detailed in this report—from data acquisition and bid stack construction to optimization and calibration—and leveraging the recommended Python-based technology stack, it is possible to develop a powerful analytical tool capable of generating deep insights into the dispatch dynamics and price formation mechanisms of the complex and evolving Japanese power market.

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